

REM2 Research Methods

Exercise 2 – 1: Climate risk in mortgage markets¹

Instructions: Read the paragraphs on pages 2 to 5 and answer the following questions.

1. Can you find the full text of this paper, i.e., Gete, P., et al. (2024)?
2. Where can you find the cited papers online?
3. Can you tell the quality of the cited papers?
4. Are the cited papers relevant to the research topic?
5. What is the period covered by cited papers? Are they recent?
6. What purposes does the literature review serve?
7. Why did the authors cite Freddie Mac.(2021a and 2021b) in the Data section?
8. Why did the authors cite Abadie et al.(2022) and Bertrand et al. (2004) in the Specification section?
9. Why did the authors cite Rossi (2021) and Higgins et al.(2022) in the Results section?
10. Is the length of the literature review section appropriate for a MSt dissertation?

¹ Source: Gete, P., et al. (2024). "Climate risk in mortgage markets: Evidence from Hurricanes Harvey and Irma." *Real Estate Economics* **52**(3): 660-686.

1. Literature Review

This article builds on an expanding literature that exploits the occurrence of hurricanes or other natural disasters as an exogenous shock to study effects on mortgage markets (see, for example, Berg & Schrader, 2012; Corté & Strahan, 2017; Garbarino & Guin, 2021; Morse, 2011; Ouazad & Kahn, 2022). Related literature has studied other financial and economic effects of hurricanes, like effects on commercial real estate (Addoum et al., 2024), housing prices (Ortega & Taspinar, 2018), Real Estate Investment Trusts (Rehse et al., 2019), retail businesses (Meltzer et al., 2021), bank stability (Schüwer et al., 2019), stock returns (Lanfear et al., 2019), managers' perception of disaster risk (Dessaint & Matray, 2017), fiscal costs (Deryugina, 2017), homeownership (Bleemer & van der Klaauw, 2019), and households' balance sheets (Billings et al., 2022; Deryugina et al., 2018). (Ouazad & Kahn, 2022) model distortions that the GSEs create for mortgage origination.⁵ They focus on distortions operating through lending standards or quantities, while we show effects on prices.

Our article is novel in analyzing how hurricane-induced default risk translates into market-based mortgage pricing. Our results show that existing mortgage rates in the United States do not reflect the climate risks that markets would price. This result brings a different risk dimension to (Hurst et al., 2016), who show that lack of risk-based pricing provides insurance across locations.

Another contribution of this article is to accurately estimate the increase in mortgage default probabilities caused by hurricanes. We show additional evidence of the effects of natural disasters on loan defaults (see, e.g., Kousky, Kunreuther, et al., 2020; Kousky, Palim, et al., 2020; Rossi, 2021).

This article also contributes to the housing finance literature. We contribute to this literature by estimating the differences between market-based pricing and GSE statutory g-fees. Papers like Lucas and McDonald (2010), Jeske et al. (2013), Frame et al. (2012), Elenev et al. (2016), Hurst et al. (2016), Gete and Zecchetto (2018), and Wachter (2018) have analyzed different topics related to the role and future of the GSEs. Pavlov et al. (2021) and Stanton and Wallace (2011) study how mortgage credit risk was not reflected in the prices of credit default swaps during the 2008 financial crisis, pointing out the failure of transferring credit risk to the market.

2. Data

We assemble a unique database by combining information at the security level from multiple data sources. First, we collect data of the mortgages in the CRTs reference pool from the GSEs (Freddie Mac., 2021a, 2021b).

3. Specification

We estimate the model for time windows of 15–45 days before and after the treatment date. In our estimation, we cluster the standard errors by CRT security (Abadie et al., 2022; Bertrand et al., 2004).

4. Results

Overall, following a hurricane the probability of default per hurricane increases by 0.62 pp (ignoring the slight change in the loan sample each year). These results are in line with (Rossi, 2021).

There are several reasons why this value is larger than the literature that finds losses given default of about 40% on average for outstanding loans (Higgins et al., 2022).

References:

1. Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2022). When Should You Adjust Standard Errors for Clustering?*. *Quarterly Journal of Economics*, 138(1), 1-35. <https://doi.org/10.1093/qje/qjac038>
2. Addoum, J. M., Eichholtz, P., Steiner, E., & Yönder, E. (2024). Climate change and commercial real estate: Evidence from Hurricane Sandy. *Real Estate Economics*, 52(3), 687-713. <https://doi.org/10.1111/1540-6229.12435>
3. Berg, G., & Schrader, J. (2012). Access to credit, natural disasters, and relationship lending. *Journal of Financial Intermediation*, 21(4), 549-568. <https://doi.org/10.1016/j.jfi.2012.05.003>
4. Bertrand, M., Duflo, E., & Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119(1), 249-275. <https://doi.org/10.1162/003355304772839588>
5. Billings, S. B., Gallagher, E. A., & Ricketts, L. (2022). Let the rich be flooded: The distribution of financial aid and distress after hurricane harvey. *Journal of Financial Economics*, 146(2), 797-819. <https://doi.org/10.1016/j.jfineco.2021.11.006>
6. Bleemer, Z., & van der Klaauw, W. (2019). Long-run net distributionary effects of federal disaster insurance: The case of Hurricane Katrina. *Journal of Urban Economics*, 110, 70-88. <https://doi.org/10.1016/j.jue.2019.01.005>
7. Corté, K. R., & Strahan, P. E. (2017). Tracing out capital flows: How financially integrated banks respond to natural disasters. *Journal of Financial Economics*, 125(1), 182-199. <https://doi.org/10.1016/j.jfineco.2017.04.011>
8. Deryugina, T. (2017). The Fiscal Cost of Hurricanes: Disaster Aid versus Social Insurance. *American Economic Journal-Economic Policy*, 9(3), 168-198. <https://doi.org/10.1257/pol.20140296>
9. Deryugina, T., Kawano, L., & Levitt, S. (2018). The Economic Impact of Hurricane Katrina on Its Victims: Evidence from Individual Tax Returns. *American Economic Journal-Applied Economics*, 10(2), 202-233. <https://doi.org/10.1257/app.20160307>

10. Dessaint, O., & Matray, A. (2017). Do managers overreact to salient risks? Evidence from hurricane strikes. *Journal of Financial Economics*, 126(1), 97-121. <https://doi.org/10.1016/j.jfineco.2017.07.002>
11. Elenev, V., Landvoigt, T., & Van Nieuwerburgh, S. (2016). Phasing out the GSEs. *Journal of Monetary Economics*, 81, 111-132. <https://doi.org/10.1016/j.jmoneco.2016.06.003>
12. Frame, W. S., Wall, L. D., & White, L. J. (2012). The devil's in the tail: Residential mortgage finance and the U.S. Treasury In A. Federal Reserve Bank of Atlanta, GA (Ed.), *Working Paper No. 2012-12*.
13. Freddie Mac. (2021a). *Credit Risk Transfer*. <https://crt.freddiemac.com/>
14. Freddie Mac. (2021b). *Freddie Mac, 30-year fixed rate mortgage average in the United States*. Retrieved from FRED,
15. *Federal Reserve Bank of St. Louis*. <https://fred.stlouisfed.org/series/MORTGAGE30US>
16. Garbarino, N., & Guin, B. (2021). High water, no marks? Biased lending after extreme weather. *Journal of Financial Stability*, 54, Article 100874. <https://doi.org/10.1016/j.jfs.2021.100874>
17. Gete, P., & Zecchetto, F. (2018). Distributional Implications of Government Guarantees in Mortgage Markets. *Review of Financial Studies*, 31(3), 1064-1097. <https://doi.org/10.1093/rfs/hhx083>
18. Higgins, E., Yavas, A., & Zhu, S. (2022). Private mortgage securitization and loss given default. *Real Estate Economics*, 50(5), 1334-1359. <https://doi.org/10.1111/1540-6229.12376>
19. Hurst, E., Keys, B. J., Seru, A., & Vavra, J. (2016). Regional Redistribution through the US Mortgage Market. *American Economic Review*, 106(10), 2982-3028. <https://doi.org/10.1257/aer.20151052>
20. Jeske, K., Krueger, D., & Mitman, K. (2013). Housing, mortgage bailout guarantees and the macro economy. *Journal of Monetary Economics*, 60(8), 917-935. <https://doi.org/10.1016/j.jmoneco.2013.09.001>
21. Kousky, C., Kunreuther, H., LaCour-Little, M., & Wachter, S. (2020). Flood Risk and the U.S. Housing Market. *Journal of Housing Research*, 29(sup1), S3-S24. <https://doi.org/10.1080/10527001.2020.1836915>
22. Kousky, C., Palim, M., & Pan, Y. (2020). Flood Damage and Mortgage Credit Risk: A Case Study of Hurricane Harvey. *Journal of Housing Research*, 29(sup1), S86-S120. <https://doi.org/10.1080/10527001.2020.1840131>
23. Lanfear, M. G., Lioui, A., & Siebert, M. G. (2019). Market anomalies and disaster risk: Evidence from extreme weather events. *Journal of Financial Markets*, 46, Article 100477. <https://doi.org/10.1016/j.finmar.2018.10.003>
24. Lucas, D., & McDonald, R. (2010). 131Valuing Government Guarantees: Fannie and Freddie Revisited. In D. Lucas (Ed.), *Measuring and Managing Federal Financial Risk* (pp. 0). University of Chicago Press. <https://doi.org/10.7208/chicago/9780226496597.003.0007>
25. Meltzer, R., Ellen, I. G., & Li, X. D. (2021). Localized commercial effects from natural disasters: The case of Hurricane Sandy and New York City. *Regional Science and Urban Economics*, 86, Article 103608. <https://doi.org/10.1016/j.regsciurbeco.2020.103608>

26. Morse, A. (2011). Payday lenders: Heroes or villains? *Journal of Financial Economics*, 102(1), 28-44. <https://doi.org/10.1016/j.jfineco.2011.03.022>
27. Ortega, F., & Taspinar, S. (2018). Rising sea levels and sinking property values: Hurricane Sandy and New York's housing market. *Journal of Urban Economics*, 106, 81-100. <https://doi.org/10.1016/j.jue.2018.06.005>
28. Ouazad, A., & Kahn, M. E. (2022). Mortgage Finance and Climate Change: Securitization Dynamics in the Aftermath of Natural Disasters. *Review of Financial Studies*, 35(8), 3617-3665. <https://doi.org/10.1093/rfs/hhab124>
29. Pavlov, A., Schwartz, E., & Wachter, S. (2021). Price Discovery Limits in the Credit Default Swap Market in the Financial Crisis. *Journal of Real Estate Finance and Economics*, 62(2), 165-186. <https://doi.org/10.1007/s11146-020-09747-8>
30. Rehse, D., Riordan, R., Rottke, N., & Zietz, J. (2019). The effects of uncertainty on market liquidity: Evidence from Hurricane Sandy. *Journal of Financial Economics*, 134(2), 318-332. <https://doi.org/10.1016/j.jfineco.2019.04.006>
31. Rossi, C. V. (2021). Assessing the impact of hurricane frequency and intensity on mortgage delinquency. *Journal of Risk Management in Financial Institutions*, 14(4), 426-442. <https://ideas.repec.org/a/aza/rmfi00/y2021v14i4p426-442.html> (Journal of Risk Management in Financial Institutions)
32. Schüwer, U., Lambert, C., & Noth, F. (2019). How Do Banks React to Catastrophic Events? Evidence from Hurricane Katrina. *Review of Finance*, 23(1), 75-116. <https://doi.org/10.1093/rof/rfy010>
33. Stanton, R., & Wallace, N. (2011). The Bear's Lair: Index Credit Default Swaps and the Subprime Mortgage Crisis. *Review of Financial Studies*, 24(10), 3250-3280. <https://doi.org/10.1093/rfs/hhr073>
34. Wachter, S. M. (2018). Credit risk transfer, informed markets, and securitization. *Economic Policy Review*(24-3), 117-137. <https://ideas.repec.org/a/fip/fednep/00057.html> (Economic Policy Review)